WEB BASED APPLICATION DEVELOPMENT (CSF3023)

Lab Report 2

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2.9 Exercise

Objective: Develop a **Product Inventory System** by applying knowledge of Servlets, JavaBeans, and the DAO pattern to create a simple dynamic web application.

Instruction

Please complete this exercise by fulfilling the following requirements:

1. **Database Setup:** Create a new table named `products` in your `CSE3023` database with the following columns:
   - `id` (INT, Primary Key, Auto Increment)
   - `name` (VARCHAR)
   - `category` (VARCHAR)
   - `price` (DOUBLE)
   - `quantity` (INT)
2. **JavaBean (Model):** Create a class named `Product` in the `com.lab.model` package. Ensure it has the appropriate private attributes, constructors (empty and parameterized), and getter/setter methods.
3. **Data Access Object (DAO):** Create a `ProductDAO` class in the `com.lab.dao` package to handle all SQL operations (`INSERT`, `SELECT`, `UPDATE`, `DELETE`).
4. **Servlets (Controller):** Create the necessary Servlets (e.g., `InsertProductServlet`, `ViewProductServlet`, `UpdateProductServlet`, `DeleteProductServlet`) to capture user inputs and interact with your `ProductDAO`. **Strict rule:** No JDBC code should be written inside these Servlets!
5. **User Interface (View):** Create an HTML form (`add_product.html`) to insert new products. Ensure your view servlet generates an HTML table displaying all products, along with functional "Edit" and "Delete" hyperlinks for each row.

index.html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Product Inventory System</title>
</head>
<body>
    <h1>Product Inventory System</h1>
    <a href="add_product.html">Add Product</a><br>
    <a href="viewProducts">View Products</a>
</body>
</html>
```

add_product.html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Add Product</title>
</head>
<body>
    <h1>Add New Product</h1>
    <form action="insertProduct" method="post">
        Name: <input type="text" name="name" required><br>
        Category: <input type="text" name="category" required><br>
        Price: <input type="text" name="price" required><br>
        Quantity: <input type="text" name="quantity" required><br>
        <input type="submit" value="Add Product">
    </form>
    <br>
    <a href="index.html">Home</a>
</body>
</html>
```

ProductDAO.java

```java
    private String url = "jdbc:mysql://localhost:3307/CSE3023";
    private String username = "root";
    private String password = "admin";

    private Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            e.printStackTrace();
        }
        return DriverManager.getConnection(url, username, password);
    }

    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    public List<Product> getAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection();
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery(sql)) {
```

```java
            while (rs.next()) {
                Product p = new Product();
                p.setId(rs.getInt("id"));
                p.setName(rs.getString("name"));
                p.setCategory(rs.getString("category"));
                p.setPrice(rs.getDouble("price"));
                p.setQuantity(rs.getInt("quantity"));
                products.add(p);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return products;
    }

    public Product getProductById(int id) {
        Product product = null;
        String sql = "SELECT * FROM products WHERE id = ?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                product = new Product();
                product.setId(rs.getInt("id"));
                product.setName(rs.getString("name"));
                product.setCategory(rs.getString("category"));
                product.setPrice(rs.getDouble("price"));
                product.setQuantity(rs.getInt("quantity"));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
```

```java
        return product;
    }

    public void updateProduct(Product product) {
        String sql = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    public void deleteProduct(int id) {
        String sql = "DELETE FROM products WHERE id=?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

Product.java

```java
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    public Product() {}

    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getCategory() { return category; }
    public void setCategory(String category) { this.category = category; }
    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }
    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
```

DeleteProductServlet.java

```java
package com.lab.servlet;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class DeleteProductServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        ProductDAO dao = new ProductDAO();
        dao.deleteProduct(id);

        response.sendRedirect("viewProducts");
    }
}
```

EditProductServlet.java

```java
public class EditProductServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        ProductDAO dao = new ProductDAO();
        Product product = dao.getProductById(id);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<head><title>Edit Product</title></head>");
        out.println("<body>");
        out.println("<h1>Edit Product</h1>");
        out.println("<form action='updateProduct' method='post'>");
        out.println("<input type='hidden' name='id' value='" + product.getId() + "'>");
        out.println("Name: <input type='text' name='name' value='" + product.getName() + "'><br>");
        out.println("Category: <input type='text' name='category' value='" + product.getCategory() + "'><br
        out.println("Price: <input type='text' name='price' value='" + product.getPrice() + "'><br>");
        out.println("Quantity: <input type='text' name='quantity' value='" + product.getQuantity() + "'><br
        out.println("<input type='submit' value='Update'>");
        out.println("</form>");
        out.println("<br><a href='viewProducts'>Back to List</a>");
        out.println("</body>");
        out.println("</html>");
```

InsertProductServlet.java

```java
package com.lab.servlet;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class InsertProductServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(name, category, price, quantity);
        ProductDAO dao = new ProductDAO();
        dao.insertProduct(product);

        response.sendRedirect("viewProducts");
```

UpdateProductServlet.java

```java
package com.lab.servlet;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class UpdateProductServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);
        ProductDAO dao = new ProductDAO();
        dao.updateProduct(product);

        response.sendRedirect("viewProducts");
```

ViewProductServlet.java

```java
public class ViewProductServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        ProductDAO dao = new ProductDAO();
        List<Product> products = dao.getAllProducts();

        out.println("<html>");
        out.println("<head><title>Product List</title></head>");
        out.println("<body>");
        out.println("<h1>Product List</h1>");
        out.println("<table border='1'>");
        out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actio

        for (Product p : products) {
            out.println("<tr>");
            out.println("<td>" + p.getId() + "</td>");
            out.println("<td>" + p.getName() + "</td>");
            out.println("<td>" + p.getCategory() + "</td>");
            out.println("<td>" + p.getPrice() + "</td>");
            out.println("<td>" + p.getQuantity() + "</td>");
            out.println("<td>");
            out.println("<a href='editProduct?id=" + p.getId() + "'>Edit</a> | ");
            out.println("<a href='deleteProduct?id=" + p.getId() + "' onclick='return confirm(\"Are you su
            out.println("</td>");
            out.println("</tr>");
```

```java
        out.println("</table>");
        out.println("<br><a href='add_product.html'>Add New Product</a>");
        out.println("<br><a href='index.html'>Home</a>");
        out.println("</body>");
        out.println("</html>");
```

Output

localhost:8080/LabExercise/index.html

Product Inventory System

Add Product
View Products

localhost:8080/LabExercise/add_product.html

Add New Product

Name: Coca cola
Category: Water
Price: 5.00
Quantity: 10
Add Product

Home

localhost:8080/LabExercise/viewProducts

Product List

ID	Name	Category	Price	Quantity	Actions
3	Coca cola	Water	5.0	10	Edit | Delete
6	Sprite	Water	1.2	10	Edit | Delete
7	Pizza	Food	6.0	15	Edit | Delete

Add New Product
Home

When Delete

Product List

ID	Name	Category	Price	Quantity	Actions
3	Coca cola	Water	5.0	10	Edit | Delete
7	Pizza	Food	6.0	15	Edit | Delete

Add New Product
Home

When Edited

Edit Product

Name: Coca cola
Category: Water
Price: 5.0
Quantity: 20
Update

Back to List

Updated

Product List

ID	Name	Category	Price	Quantity	Actions
3	Coca cola	Water	5.0	20	Edit | Delete
7	Pizza	Food	6.0	15	Edit | Delete

Add New Product
Home